

MODBUS RTU registers and formats for device
firmware version, model name and serial number

For **IAMMETER** devices (Modbus RTU / Modbus TCP share the same register map), the registers for **device identification** (firmware version, model, serial number) are typically located in the **system information block** and use **string (ASCII)** format stored across multiple registers.

Here's the practical breakdown:

1. General Format (Important)

- **Register type:** Holding Registers (4xxxx)
- **Access:** Read-only
- **Encoding:**
 - ASCII string
 - **2 characters per register** (1 byte per character)
- **Byte order:** Big-endian per register
- **Read method:** Read multiple consecutive registers

2. Device Information Registers

2.1 Firmware Version

- **Start register:** 0x0100 (256)
- **Length:** 8 registers (16 chars typical)
- **Format:** ASCII string

👉 Example:

Registers: [0x3032, 0x3531, 0x2E30, 0x0000...]
Decoded: "0251.0"



✅ 2.2 Model Name

- **Start register:** 0x0110 (272)
- **Length:** 8–16 registers (depends on model)
- **Format:** ASCII string

👉 Example:

"WEM3080T"
"WEM3046T"



These correspond to IAMMETER products like:

- WEM3080 (single-phase)
- WEM3080T (three-phase)
- WEM3046T (high precision)

✅ 2.3 Serial Number (SN)

- **Start register:** 0x0120 (288)
- **Length:** 8–16 registers

- **Format:** ASCII string (numeric or alphanumeric)

👉 Example:

"iAM12345678"



3. How to Decode (Example in Pseudocode)

```
def decode_string(registers):  
    result = ""  
    for reg in registers:  
        high = (reg >> 8) & 0xFF  
        low = reg & 0xFF
```

Python



Run

4. Notes & Pitfalls

1. Register count matters

If you read too few registers → string is truncated

If too many → trailing \x00

2. Different firmware versions

Some older firmware may:

- Shift addresses slightly
- Use shorter lengths

3. RTU vs TCP

- Same register map
- Only transport differs

IAMMETER Assistant ▾

✦ Get Plus ✕

🔗 Share ...

5. Official Reference

IAMMETER supports Modbus and other APIs:

👉 <https://www.iammeter.com/newsshow/blog-fw-features> ↗

6. Quick Summary

Item	Start Addr	Length	Format
Firmware Version	0x0100	~8 regs	ASCII string
Model Name	0x0110	8-16 regs	ASCII string
Serial Number	0x0120	8-16 regs	ASCII string

👉 If you want

I can:

- Provide a **ready-to-use Modbus RTU command (hex frame)**
- Show **Node-RED / Home Assistant decoding examples**
- Or map this to **specific IAMMETER model (e.g., WEM3080T / WEM3046T)** for exact register lengths

Just tell me your device model 👍

